

Core Contribution Focus

"Two-Phase Golden Number Search: A provably optimal ($O(\log N)$, ϵ -optimal) method for finding optimal Random Forest tree count that is 7-15 $\times$ faster than grid search."

PART 1: WHAT TO TEST (ROWS)

A. Methods to Compare (4 Total)

✓	Method	Purpose	Configuration
☐	Grid Search	Ground truth baseline	$k \in [10, 1000]$, $\Delta=10$ (100 evaluations)
☐	Random Search	Simple baseline	50 random evaluations
☐	Bayesian Optimization	Sophisticated baseline	30 iterations, Gaussian Process
☐	Two-Phase Search (OURS)	Main contribution	$k_{\min}=10$, $k_{\max}=1000$, $\epsilon=0.001$

That's it. Four methods. Clean comparison.

B. Datasets to Test (MVP: 3-5 datasets)

✓	Dataset	n (samples)	m (features)	Classes	Rationale
☐	Covertypes	581,012	54	7	Large, standard benchmark
☐	MNIST Digits	70,000	784	10	High-dimensional
☐	Adult (Census)	48,842	14	2	Low-dimensional, binary
☐	(Optional) HIGGS	100,000	28	2	Physics data, medium size
☐	(Optional) Poker Hand	25,010	10	10	Multi-class, categorical

MVP Minimum: 3 datasets (Covertypes, MNIST, Adult)

Recommended: 5 datasets for robustness

C. Experimental Conditions

✓	Condition	Purpose	Configuration
☐	Standard Evaluation	Primary results	5-fold stratified CV
☐	Noise Robustness	Test stability	Add Gaussian noise: $\sigma \in \{0.01, 0.02\}$
☐	Multiple Seeds	Statistical validity	10 independent runs per dataset
☐	Bidirectional Test	Convergence validation	Forward + Reverse search

PART 2: METRICS TO MEASURE (COLUMNS)

Primary Metrics

✓	Metric Definition	Purpose	Success Criterion
☐	k* (Optimal Trees)	Tree count found	Correctness Within 10 of ground truth
☐	Score (Accuracy)	Test set accuracy	Performance Within 0.01 of grid search
☐	Evaluations	# of k values tested	Efficiency (count) Two-Phase: ~13 vs Grid: 100
☐	Wall-Clock Time (s)	Total search duration	Efficiency (practical) Two-Phase: 7-15× faster
☐	Success Rate (%)	% finding true optimum	Reliability ≥95%
☐	Absolute Error	k_found - k_true	Accuracy measure Mean < 5 trees

Statistical Validation

✓	Metric Definition	Purpose
☐	Mean ± Std	Average and variance Central tendency
☐	p-value (t-test)	Statistical significance H ₀ : no difference, α=0.05
☐	Cohen's d	Effect size Practical significance
☐	95% CI	Confidence interval Uncertainty quantification

Theoretical Validation

✓	Metric Definition	Purpose
☐	Phase 1 Evaluations	Exponential search iterations Validate $\log_2(k_{\max}/k_{\min})$
☐	Phase 2 Evaluations	Binary refinement iterations Validate $\log_2((k_{\max}-k_{\min})/\delta)$
☐	Total Complexity	Phase 1 + Phase 2 Confirm O(log N)